

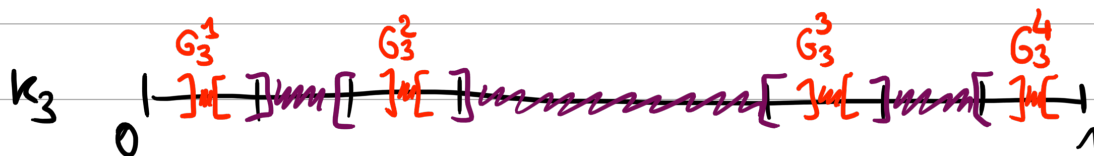
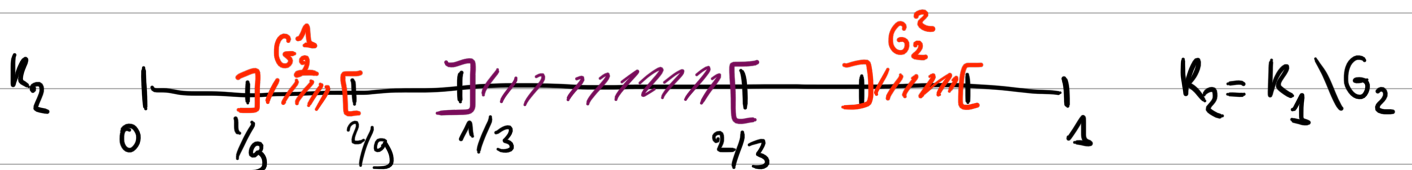
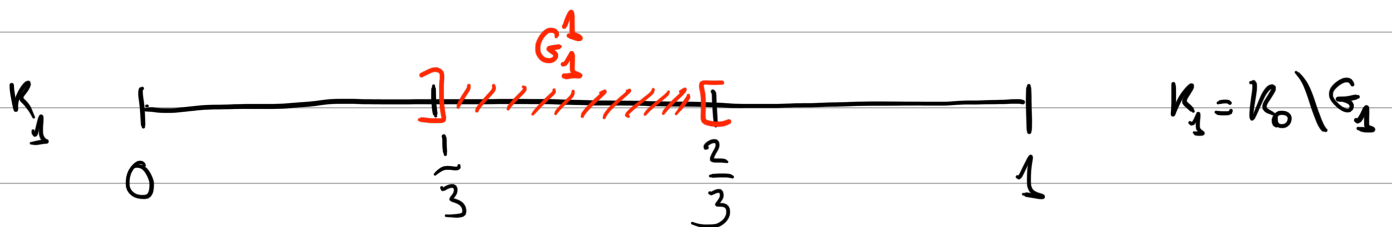
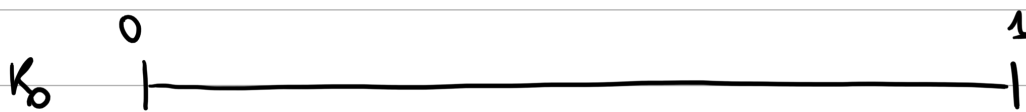
15.12.2022

Def 2 (Cantor set)

$$K = \bigcap_{n \geq 1} K_n$$

$$K_0 = [0, 1]$$

K_{n+1} is obtained after removing the open middle third of each interval composing K_n



We write $K_n = K_{n-1} \setminus G_n$

where $G_n = G_n^1 \cup \dots \cup G_n^{2^{n-1}}$ is composed of 2^{n-1} intervals of length $\frac{1}{3^n}$. [disjoint]

Since $K = \bigcap_{n \geq 1} K_n = [0, 1] \setminus \left(\bigcup_{n \geq 1} G_n \right)$:

- K is closed because all the intervals G_n^i are open so K is a Borel set

• Therefore

$$\lambda(K) = 1 - \lambda\left(\bigcup_{n \geq 1} G_n\right)$$

where

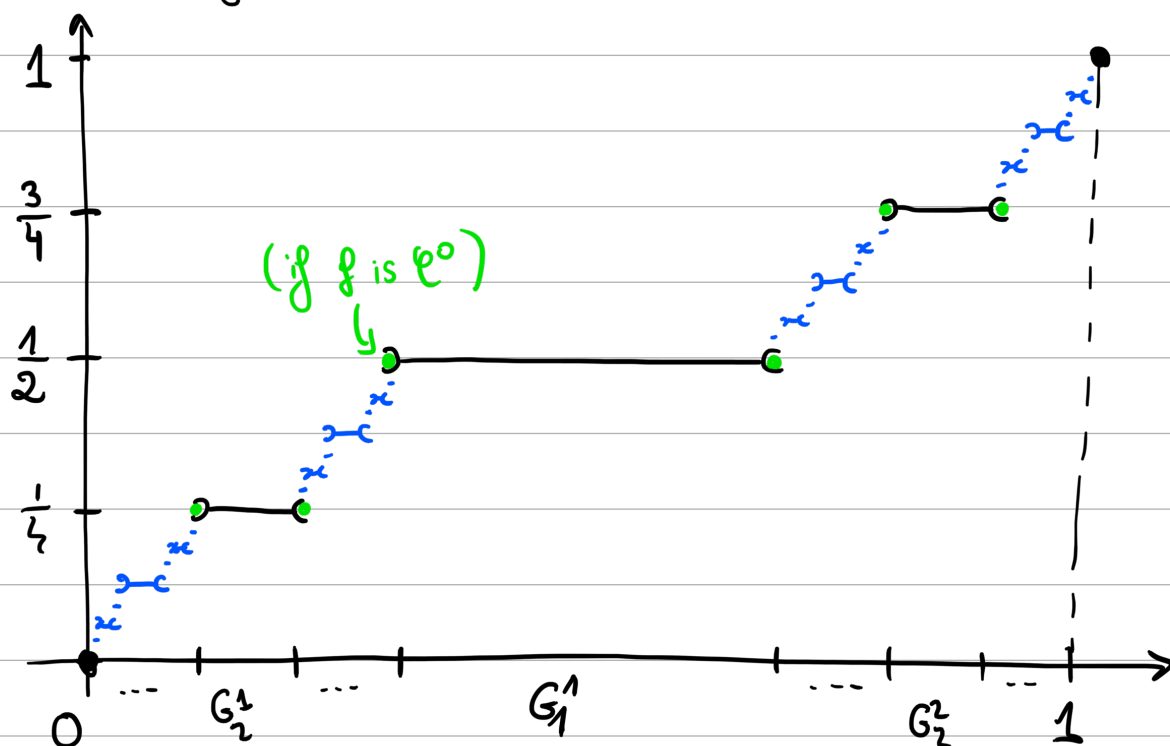
$$\lambda\left(\bigcup_{n \geq 1} G_n\right) = \sum_{n \geq 1} \lambda(G_n) = \sum_{n \geq 1} \frac{2^{n-1}}{3^n}$$

$$\text{Since } \sum_{n \geq 1} \frac{2^{n-1}}{3^n} = \frac{1}{2} \sum_{n \geq 1} \left(\frac{2}{3}\right)^n = \frac{1}{2} \frac{\frac{2}{3}}{1 - \frac{2}{3}} = 1,$$

we deduce that $\lambda(K) = 0$.

Since $\lambda(K) = 0$, K cannot contain an interval that has more than one element.

Def 3 (Cantor function)



We want to define

$$f(x) = \frac{1}{2} \text{ on } G_1^1$$

$$f(x) = \frac{1}{4} \text{ on } G_2^1$$

...

$$f(x) = \frac{3}{4} \text{ on } G_2^2$$

$$f(x) \in \left\{ \frac{k}{2^n} \mid k \in \mathbb{N} \right\} \setminus \left\{ \frac{k}{2^{n-1}} \mid k \in \mathbb{N} \right\} \text{ on } G_n^i$$

Then f would be defined on $\bigcup_{n \geq 1} G_n$, and we can set

$$f(0) = 0$$

$$f(x) = \sup \{ f(t) \mid t \in [0, 1] \setminus K \text{ and } t < x \}$$

Prop 4 The Cantor function f is nondecreasing, continuous
 $\lfloor$ and $f(K) = [0, 1]$

Proof

• We use base 3 representation to describe K .

We write every $x \in [0, 1]$ as $x = \sum_{i \geq 1} a_i \left(\frac{1}{3}\right)^i$, $a_i \in \{0, 1, 2\}$.

Then G_1 is the set of $x \in [0, 1]$ such that $a_1 = 1$ in any base-3 representation.

Similarly, G_2 is the set of $x \in [0, 1]$ such that $a_1 \in \{0, 2\}$ and $a_2 = 1$ in any base-3 representation

...

G_n is the set of $x \in [0, 1]$ such that $a_1, \dots, a_{n-1} \in \{0, 2\}$ and $a_n = 1$ in any base-3 representation

So K is the set of $x \in [0, 1]$ such that in one base-3 representation, $a_i \in \{0, 2\}$.

[Ex] $\lfloor$ $\frac{1}{3} = \underset{a_1=1}{1} \times \frac{1}{3} = \underset{a_i=2 \text{ for every } i \geq 2}{2} \left(\frac{1}{3}\right)^2 + 2 \left(\frac{1}{3}\right)^3 + \dots \in K$

• Cantor function and base-3 representation:

If $x \in G_n$, we write $x = \sum_{i \geq 1} a_i \left(\frac{1}{3}\right)^i$ where
 $a_i \in \{0, 2\}$ for $i \leq n-1$ and $a_n = 1$

We define $f(x) := \sum_{i=1}^{n-1} b_i \left(\frac{1}{2^i}\right) + \frac{1}{2^n}$

where $b_i = \begin{cases} 0 & \text{if } a_i = 0 \\ 1 & \text{if } a_i = 2 \end{cases}$.

→ if $x \in G_1^1$, $a_1 = 1$ so $f(x) = \frac{1}{2}$

→ if $x \in G_2^1$, $a_1 = 0$ and $a_2 = 1$ so $f(x) = \frac{1}{4}$

if $x \in G_2^2$, $a_1 = 2$ and $a_2 = 1$ so $f(x) = 1 \times \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$

→ ...

If $x \in K$, we define $f(x) := \sum_{i \geq 1} b_i \left(\frac{1}{2}\right)^i$ where $b_i = \begin{cases} 0 & \text{if } a_i = 0 \\ 1 & \text{if } a_i = 2 \end{cases}$.

• let us show that $f(K) = [0, 1]$.

First, $0 \leq \sum_{i \geq 1} b_i \frac{1}{2^i} \leq \sum_{i \geq 1} \frac{1}{2^i} = 1$: $f(K) \subseteq [0, 1]$.

let $y \in [0, 1]$. The base 2 representation of y is of the form
 $y = \sum_{i \geq 1} b_i \frac{1}{2^i}$.

Choosing $a_i = \begin{cases} 0 & \text{if } b_i = 0 \\ 2 & \text{if } b_i = 1 \end{cases}$ and $x = \sum_{i \geq 1} a_i \frac{1}{3^i}$,

we see that $x \in K$ and $f(x) = y$.

• let us show that f is nondecreasing.

let $x, x' \in [0, 1]$ such that $x < x'$.

We write $x = \sum_{i \geq 1} a_i \frac{1}{3^i}$, $x' = \sum_{i \geq 1} a'_i \frac{1}{3^i}$ where $a_i, a'_i \in \{0, 1, 2\}$.

There exists n such that $a_1 = a'_1, \dots, a_{n-1} = a'_{n-1}$ but $a_n < a'_n$.

Case 1 if $x \notin K$ and there exists $m < n$ such that $a_m = 1$.

In this case $a'_m = 1$ and

$$f(x) = f(x') = \sum_{i=1}^{m-1} b_i \frac{1}{2^i} + \frac{1}{2^m}$$

Case 2 Otherwise, for every $m < n$, $a_m \in \{0, 2\}$.

Case 2a If $a_n = 0$ and $a'_n = 2$:

$$(1) \quad f(x) \leq \sum_{i=1}^{n-1} b_i \frac{1}{2^i} + 0 + \sum_{i=n+1}^{+\infty} 1 \frac{1}{2^i} = \sum_{i=1}^{n-1} b_i \frac{1}{2^i} + \frac{1}{2^n}.$$

$$(2) \quad f(x') \geq \sum_{i=1}^{n-1} b_i \frac{1}{2^i} + \frac{1}{2^n} + \sum_{i=n+1}^{+\infty} 0 \cdot \frac{1}{2^i} = \sum_{i=1}^{n-1} b_i \frac{1}{2^i} + \frac{1}{2^n}.$$

so $f(x) \leq f(x')$.

Case 2b If $a_n = 0$ and $a'_n = 1$:

using (1),
$$f(x) \leq \sum_{i=1}^{n-1} b_i \frac{1}{2^i} + \frac{1}{2^n}$$

$$\text{and } f(x') = \sum_{i=1}^{n-1} b_i \frac{1}{2^i} + \frac{1}{2^n}$$

so $f(x) \leq f(x')$.

Case 2c If $a_n = 1$ and $a'_n = 2$:

using (2),
$$f(x') \geq \sum_{i=1}^{n-1} b_i \frac{1}{2^i} + \frac{1}{2^n}$$

$$\text{and } f(x) = \sum_{i=1}^{n-1} b_i \frac{1}{2^i} + \frac{1}{2^n}$$

so $f(x) \leq f(x')$.

- f is continuous because it is nondecreasing and $f([0, 1]) \supseteq f(K) = [0, 1]$.